

***B.Tech. Degree II Semester Regular and Supplementary/  
I Semester Supplementary Examination April 2018***

**CE/EE/ME/SE AS 15 – 1202 A & CS/EC/IT AS 15 – 1102 B  
ENGINEERING CHEMISTRY  
(2015 Scheme)**

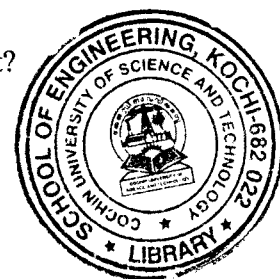
Time : 3 Hours

Maximum Marks : 60

**PART A**  
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain shielding in NMR spectroscopy.
- (b) Write any four methods to improve the reactivity in solids.
- (c) Derive the relation between electrode potential and concentration of electrolyte.
- (d) What is Overvoltage? What are the factors influencing Overvoltage?
- (e) What is pseudo order reaction? Explain with example.
- (f) What are the different statements of 2<sup>nd</sup> law of Thermodynamics?
- (g) What are the important chemical reactions involved in the setting of cement?
- (h) Give the importance of viscosity index in the efficiency of lubricants.
- (i) Differentiate Galvanisation and Thinning.
- (j) What is vulcanization? Give its importance in Rubber Technology.



**PART B**

(4 × 10 = 40)

- II. (a) Explain BET theorem. Describe the thermodynamic importance of adsorption. (2½)
- (b) What is Beer-Lambert's law? (1½)
- (c) Draw the splitting of signals in NMR spectra of (6)
- (i) CH<sub>3</sub>-CH<sub>2</sub>-F (ii) CF<sub>2</sub>-CH<sub>2</sub>-CF<sub>2</sub> (iii) CF<sub>2</sub>-CF<sub>2</sub>
- (iv) CHF<sub>2</sub>-CH<sub>2</sub>F

**OR**

- III. (a) What are the postulates of free electron theory? (3)
- (b) What is Born-Haber cycle? Explain with example. (3½)
- (c) Explain the behavior of different classes of solids based on band theory. (3½)
- IV. (a) Derive the integral rate law expression for first law of thermodynamics. (2½)
- (b) What are the different coating techniques in paint technology? (4)
- (c) What are the special features of concentration cell? Derive the EMF of electrode concentration cell. (3½)

**OR**

- V. Write notes on the following. (10)
- (i) Polarisation.
- (ii) Photovoltaic effect-working of solar cell.
- (iii) Buffer solution and its working with example.
- (iv) Cathodic protection.

(P.T.O.)

- VI. (a) Using phase rule explain phase diagram of water system. (4)  
(b) Derive the Kirchoff's equations. (3)  
(c) Explain variation of Gibb's free energy with temperature and pressure. (3)

**OR**

- VII. (a) What is Eutectic system? Explain the phase diagram of Ph-Ag system. (4)  
(b) Explain the following: (i) Trauton's Rule (ii) Nernst Heat theorem. (3)  
(c) Derive the Gibb's-Hemholz equation. (3)

- VIII. (a) What are Refractories? Explain important properties of Refractories. (3)  
(b) Differentiate injection and extrusion molding. (4)  
(c) Differentiate thermoplastics and thermosetts. (3)

**OR**

- IX. (a) Explain the preparation of polystyrene with Ziegler-Natta Coordination technique. (4)  
(b) What are the different theories of lubrication? (4)  
(c) Give one method of preparation of PET and PVC. (2)

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